

Neset Kablan

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EDUCATION

Northeastern University

Bachelor of Science in Electrical and Computer Engineering

Boston, MA

Sep. 2026 – Expected May 2030

PROJECTS

Closed-Loop Motor Controller ASIC | Verilog, OpenLane, Tiny Tapeout

Aug 2026 – Present

- Designed a closed-loop motor controller in Verilog and taped it out on SkyWater 130 nm silicon through a multi-project wafer shuttle
- Implemented control arithmetic in fixed point, closing timing at 50 MHz after synthesis
- Verified with a self-checking testbench and gate-level simulation, then brought up the returned die on hardware

Robot Autonomous and Shooting Systems | Java, Kotlin, Limelight 3A

Sep. 2025 – May 2026

- Derived a shoot-on-the-move solver that projects robot state forward across system latency, then binary searches time of flight to solve turret heading and flywheel RPM
- Built an autonomous framework that cut 12 competition routines to 20–30 lines each, with alliance mirroring generating red field geometry from blue definitions
- Integrated Limelight AprilTag relocalization behind a staleness gate to correct odometry drift mid-match
- Modeled turret torque and flywheel spin-up to size the shooter motor and gearing

Philidor | Python, python-chess, UCI, Lichess database

Aug 2026 – Present

- Building a chess agent that plays for draws, selecting moves by highest draw probability from engine win-draw-loss evaluation
- Wrote a threaded UCI engine wrapper with timeout handling, and a board subclass with incremental Zobrist hashing for fast repetition detection, covered by unit tests
- Training a network on Lichess games with a reward function ranking draws above wins, as a research project

Artifact Tracker | Python, OpenCV, NumPy, Raspberry Pi

Aug. 2025 – May 2026

- Built a Raspberry Pi vision system that counts game elements by HSV color thresholding and blob area, with debounce to reject duplicate detections
- Wrote an interactive calibration tool that samples ball colors by click and writes thresholds, region of interest, and alliance to a configuration file
- Removed venue-specific retuning from the code path, so lighting changes need no rebuild

EXPERIENCE & LEADERSHIP

Team Captain

Aug. 2024 – June 2026

FIRST Tech Challenge Team 3747, The Hive

Sandy, UT

- Captained a 15-student team, owning software architecture, robot design, and competition strategy.
- Led the team to back-to-back-to-back appearances at the World Championship, winning two awards there.

Program Delivery Partner Assistant

June 2025 – July 2026

FIRST LEGO League Utah

Sandy, UT

- Supported statewide program delivery and competition events for participating teams

Barista

Jan. 2026 – Sep. 2026

Starbucks

Sandy, UT

- Worked 35 hours per week through a full school year while leading a robotics team and carrying a full course load

TECHNICAL SKILLS

Hardware: Verilog, digital logic design, closed-loop control, motor drivers, Raspberry Pi, Limelight vision

Software: Java, Python, TypeScript, SQL (PostgreSQL), OpenCV, NumPy, Git, Gradle, Linux

Awards: National Merit Scholar, AP Scholar with Honor, Congressional Award Gold Medal, TSA Utah First Place in Essays in Technology

Languages: English, Turkish